

1. Motivation
2. Data Source

3. Data Analysis The analytical pipeline was structured into three distinct stages to systematically evaluate the dataset:

- **Exploratory Data Analysis (EDA):** Visual distributions and box plots were generated to examine game outcomes against scheduling data, specifically focusing on how rest days affect team win rates and how travel distances are distributed across away games.
- **Statistical Hypothesis Testing:** Independent two-sample t-tests were conducted to evaluate core assumptions, including the performance penalty of back-to-back games, cross-country flights, and the highly-debated Denver altitude advantage.
- **Predictive Modeling (Machine Learning):** Logistic Regression and Random Forest Classifiers were successfully trained using the engineered features (`Distance_Traveled_km`, `Days_of_Rest`, and `Altitude_Difference`) to predict the binary game outcome (`Win`).

4. Findings The empirical results of the analysis revealed several critical insights into NBA scheduling constraints:

- **Schedule Congestion:** The statistical testing confirmed a significant performance drop during back-to-back games. Teams playing with zero days of rest showed a lower win probability compared to those with standard rest.
- **The Travel Tax:** Teams executing long-distance cross-country flights (>1500 km) experienced a statistically significant reduction in offensive scoring output compared to minimal travel scenarios (<500 km).
- **The Denver Altitude Effect:** Surprisingly, the independent t-test regarding the Denver altitude effect failed to reach statistical significance ($p > 0.05$). When controlling for overall away conditions, high altitude alone did not conclusively suppress opponent scoring in the selected sample.
- **Predictive Capabilities:** The Machine Learning models achieved moderate predictive success on the testing data, with the Random Forest Classifier slightly outperforming Logistic Regression (54.13% vs. 51.42% accuracy). Feature importance analysis highlighted that `Distance_Traveled_km` (39.17%) and `Altitude_Difference` (37.12%) held the highest predictive weight, while `Days_of_Rest` contributed 23.71%.

5. Limitations and Future Work While this project successfully identified significant trends regarding travel distances and schedule congestion, several limitations exist. First, the predictive accuracy of the machine learning models remained around 54.13%, which indicates that game outcomes are highly complex and heavily influenced by variables not captured in this dataset, such as player injuries, team tactical matchups, in-season roster changes, and individual player form.

- Future work could extend this research by integrating baseline team strength metrics, such as real-time Elo ratings or sportsbooks point spreads. Furthermore, incorporating player-specific minutes-restriction data or injury reports would allow for a more granular assessment of how travel fatigue directly impacts individual performance dips.